Tópicos Avançados em Novas Tecnologias

Primeiros passos com docker

Prof. Diego Cardoso Borda Castro

Resumo

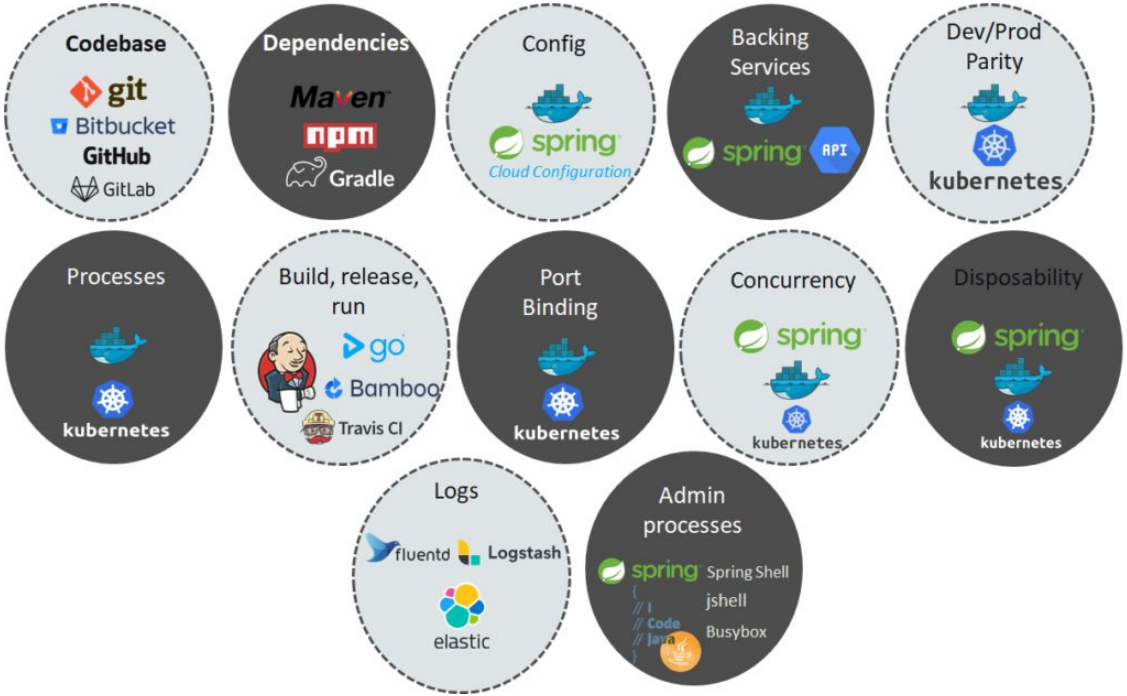
- Conceitos
  - Arquitetura
- Comandos
  - Container
  - Imagens
  - Network
  - Volumes
- Boas Práticas
- Docker Registry
- Docker Compose

Conceitos

- Desenvolvimento
  - Troca de tecnologias entre projetos
    - Instalar tudo na máquina
    - Manual de setup
- Operação
  - Subir no servidor
    - Concorrência
    - Conflito de ambiente
  - Twelve factor app

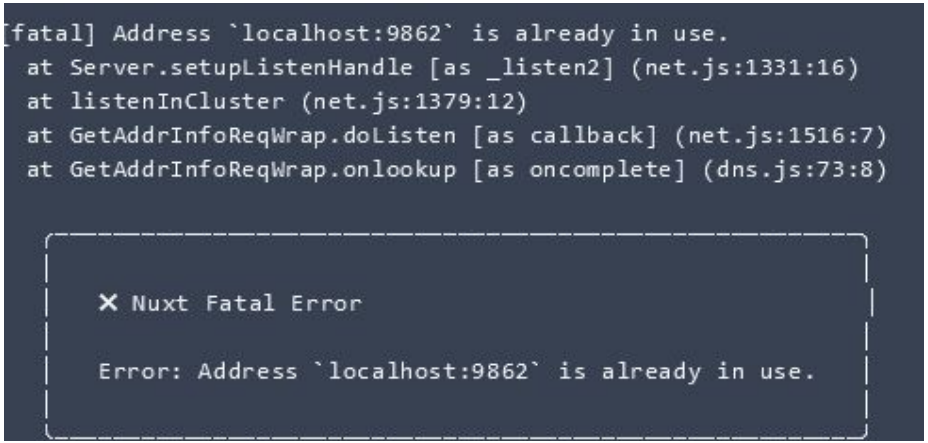
Conceitos

- Twelve factor app
  - Boas práticas



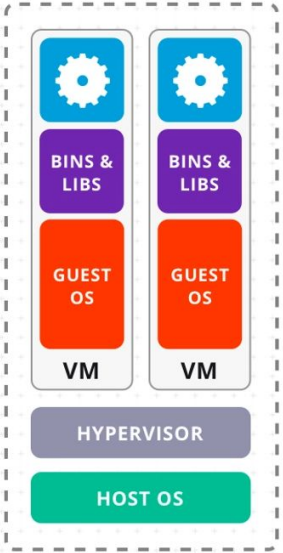
Conceitos

- Como surgiram os containers?
  - Na minha máquina funciona!
    - Concorrência
      - Projeto entregue
      - Não roda no servidor

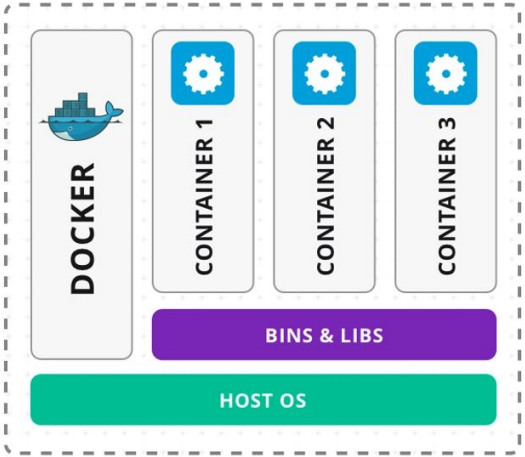


Conceitos

- Como surgiram os containers?
  - Máquinas virtuais
    - Para vários serviços, várias máquinas



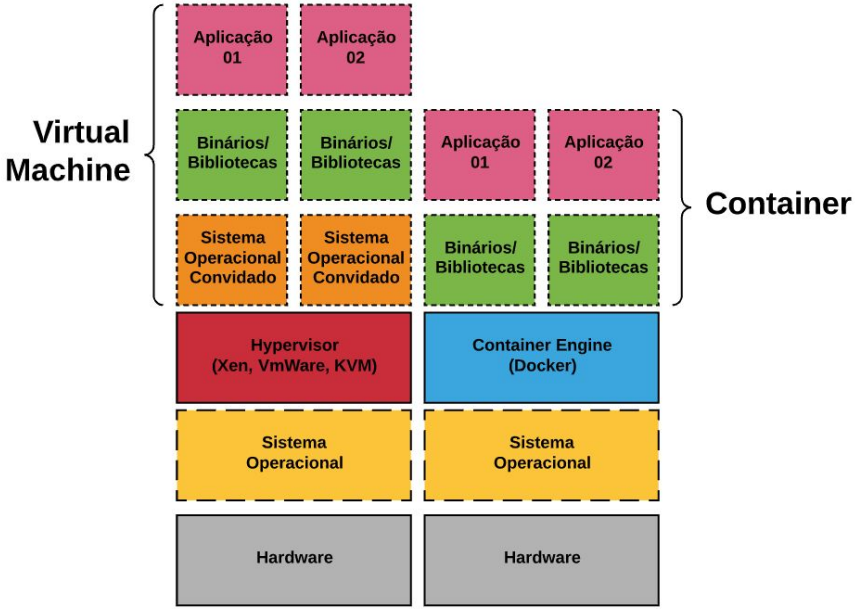
VM



Docker

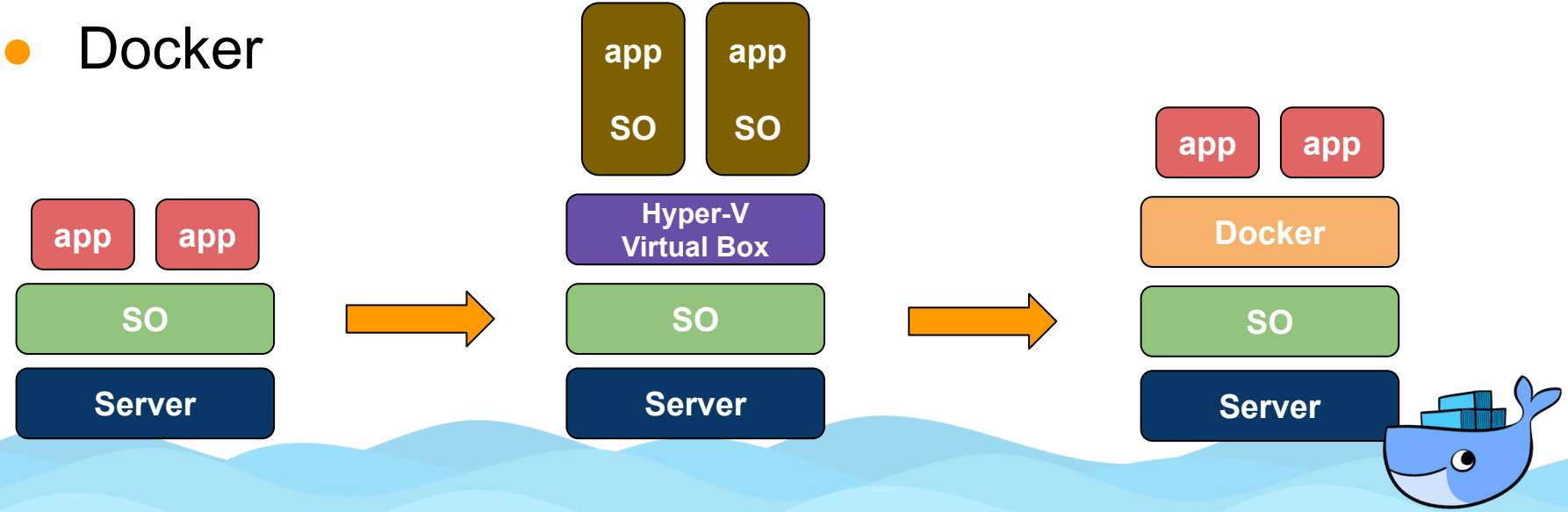
Conceitos

- Desafios VM
  - Capacidade subutilizada!
    - Uso esporádico
  - Concorrência
  - Configuração
  - Atualização
  - Segurança
  - Recursos
    - Memória, cpu, …



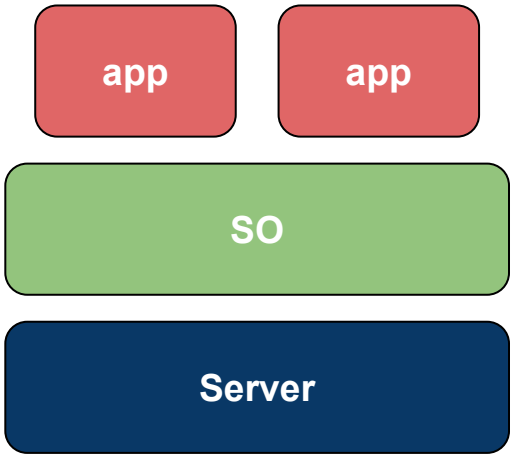
Conceitos

- Máquina física
- VM
- Docker



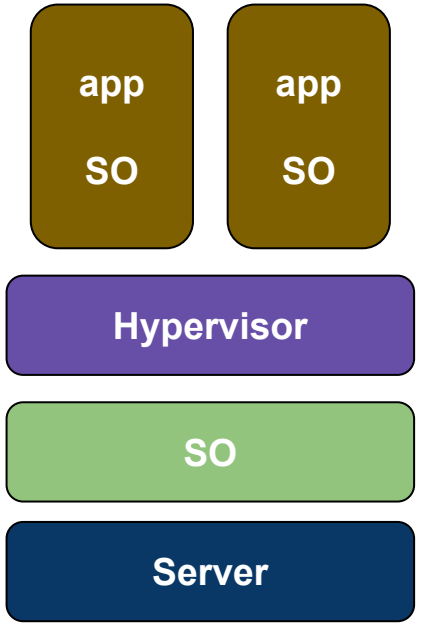
Conceitos

- Executando em uma máquina
  - Concorrência entre portas
  - Dependências entre recursos
    - CPU
    - Memória
    - Rede
    - …



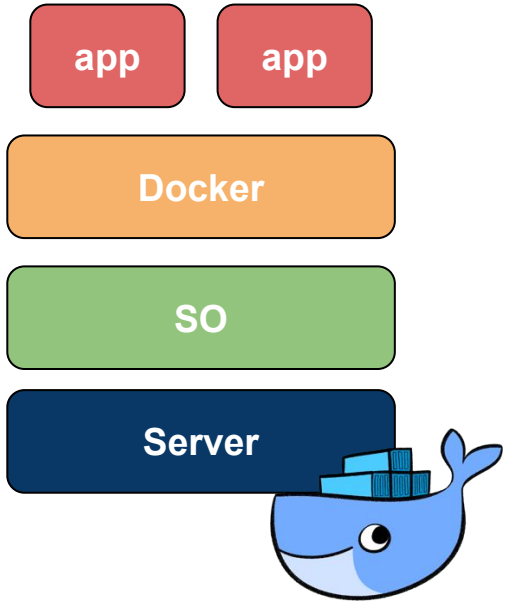
Conceitos

- Hyper-V
  - Compartilhar os recursos
  - Pequenas máquinas
  - Recursos superdimensionados



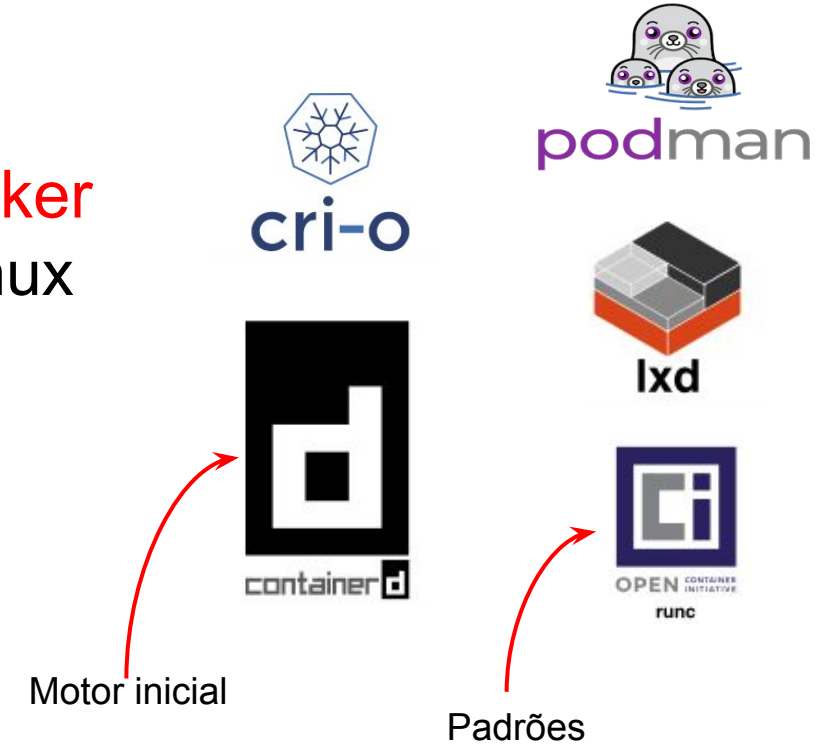
Conceitos

- Container
  - Isolamento de processos
  - Ambiente de virtualização
  - Menos processamento
  - Menor manutenção/configuração
  - Menor tempo de iniciar / derrubar
  - Facilidade de escalabilidade
  - Controle sobre os recursos



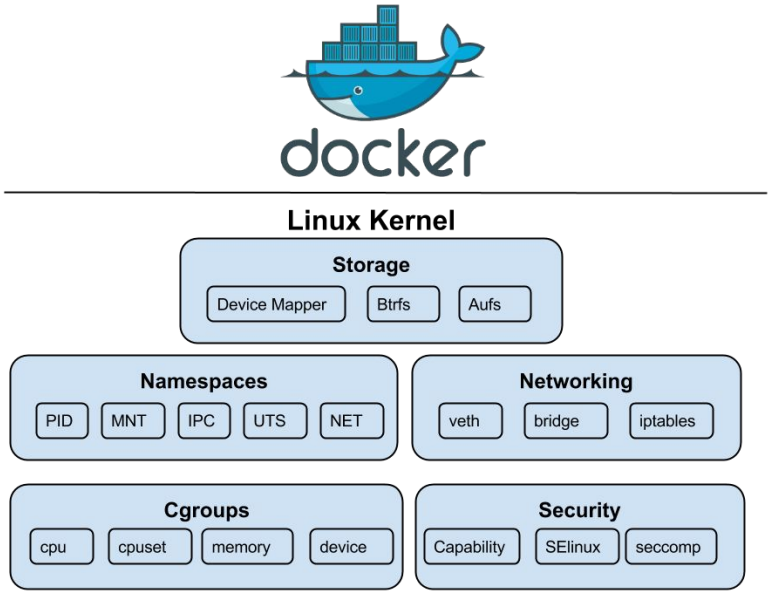
Linux

- Container
  - Não foi criado com o docker
  - Recursos do kernel do Linux
    - Namespace
      - Isolar processos
    - cgroups
      - Limitar recursos



Linux

- /boot
  - Arquivos de inicialização
- /bin
  - Principais programas
- /lib
  - Dependências
- /proc
  - Informações dos processos



Linux

```
root@diego-XPS-13-9300: /home/diego
sudo unshare --pid --fork --mount-proc /bin/bash
[sudo] senha para diego:
root@diego-XPS-13-9300:/home/diego#
```

ps -ef --forest

```
diego     460773    2560  0 18:58 ?        00:00:01  \_ /usr/local/bin/zellij --server /run/user/1000/zelli
diego     460807  460773  0 18:58 pts/4    00:00:00  |   \_ /bin/bash
root      460975  460807  0 18:59 pts/4    00:00:00  |       \_ sudo unshare --pid --fork --mount-proc /bin
root      460979  460975  0 18:59 pts/5    00:00:00  |           \_ sudo unshare --pid --fork --mount-proc
root      460980  460979  0 18:59 pts/5    00:00:00  |               \_ unshare --pid --fork --mount-proc /
root      460981  460980  0 18:59 pts/5    00:00:00  |                   \_ /bin/bash
diego     461317    2560  2 19:00 ?        00:00:01  \_ /usr/local/bin/zellij --server /run/user/1000/zelli
diego     461351  461317  0 19:00 pts/7    00:00:00      \_ /bin/bash
diego     461530  461351 99 19:01 pts/7    00:00:00          \_ ps -ef --forest
root        3626       1  0 fev24 ?        00:00:46 /usr/libexec/fwupd/fwupd
root      194214       1  0 fev28 ?        00:00:21 /usr/libexec/flatpak-system-helper
_flatpak  194603  194214  0 fev28 ?        00:00:00  \_ /usr/libexec/revokefs-fuse --backend --socket=3 --e
root      380333       1  0 mar06 ?        00:00:00 /usr/sbin/cupsd -l
```

Linux

```
pwd
/home/diego/container
sudo chroot .
chroot: failed to run command '/bin/bash': No such file or directory
ldd /bin/bash
        linux-vdso.so.1 (0x000074cce4942000)
        libtinfo.so.6 => /lib/x86_64-linux-gnu/libtinfo.so.6 (0x000074cce4784000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000074cce4400000)
        /lib64/ld-linux-x86-64.so.2 (0x000074cce4944000)
```

```
sudo apt update && sudo apt install debootstrap
[sudo] senha para diego:
Obter:1 https://repository.spotify.com stable InRelease [3.316 B]
Erro:1 https://repository.spotify.com stable InRelease
  As assinaturas a seguir não puderam ser verificadas devido à chave pública não est
Y 5384CE82BA52C83A
Atingido:2 https://updates.signal.org/desktop/apt xenial InRelease
Atingido:3 https://cli.github.com/packages stable InRelease
Ignorar:4 https://repo.vivaldi.com/stable/deb stable InRelease
```

Linux

[1]

```
~/container
sudo debootstrap jammy .
[sudo] senha para diego:
I: Retrieving InRelease
I: Checking Release signature
I: Valid Release signature (key
```

[2]

```
~/container/ubuntu-root
ls
Permissions Size User Date Modified Name
lrwxrwxrwx     - root  7 mar 19:40  bin -> usr/bin
drwxr-xr-x     - root 18 abr  2022  boot
drwxr-xr-x     - root  7 mar 19:40  dev
drwxr-xr-x     - root  7 mar 19:41  etc
drwxr-xr-x     - root 18 abr  2022  home
lrwxrwxrwx     - root  7 mar 19:40  lib -> usr/lib
lrwxrwxrwx     - root  7 mar 19:40  lib64 -> usr/lib64
drwxr-xr-x     - root  7 mar 19:40  media
drwxr-xr-x     - root  7 mar 19:40  mnt
drwxr-xr-x     - root  7 mar 19:40  opt
drwxr-xr-x     - root 18 abr  2022  proc
```

[3]

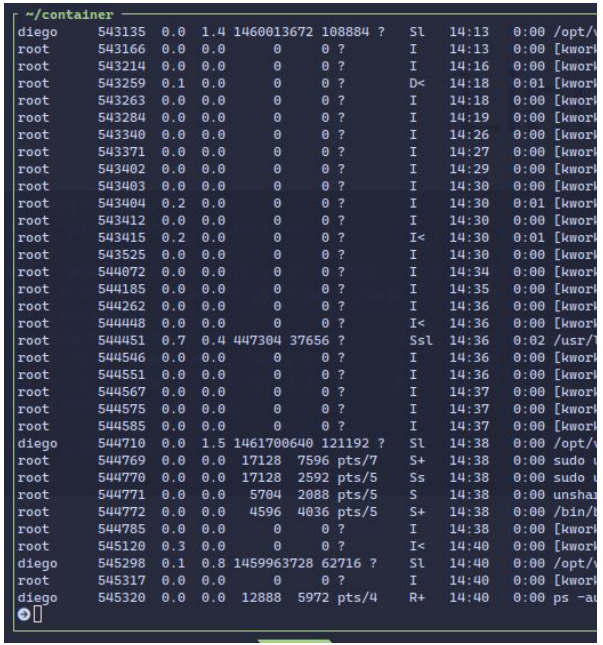
```
root@diego-XPS-13-9300: /
sudo chroot .
root@diego-XPS-13-9300:/#
```

```
~/container
root@diego-XPS-13-9300:~# cd ~
root@diego-XPS-13-9300:~# echo "teste" > teste.txt
root@diego-XPS-13-9300:~# ls
teste.txt
root@diego-XPS-13-9300:~# exit
exit
sudo ls /root/
snap
sudo ls root/
teste.txt
```

Linux

- Processos isolados
- Diretório isolados

```
root@diego-XPS-13-9300: /
sudo unshare --pid --fork --mount-proc chr
chroot  chrt
sudo unshare --pid --fork --mount-proc chroot . /bin/bash
root@diego-XPS-13-9300:/# mount -t proc proc /proc
root@diego-XPS-13-9300:/# ps -aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0   4596  4028 ?        S    17:38   0:00 /bin/bash
root           9  0.0  0.0   9200  4712 ?        R+   17:39   0:00 ps -aux
root@diego-XPS-13-9300:/# ls home/
teste.txt
root@diego-XPS-13-9300:/#
```



Linux

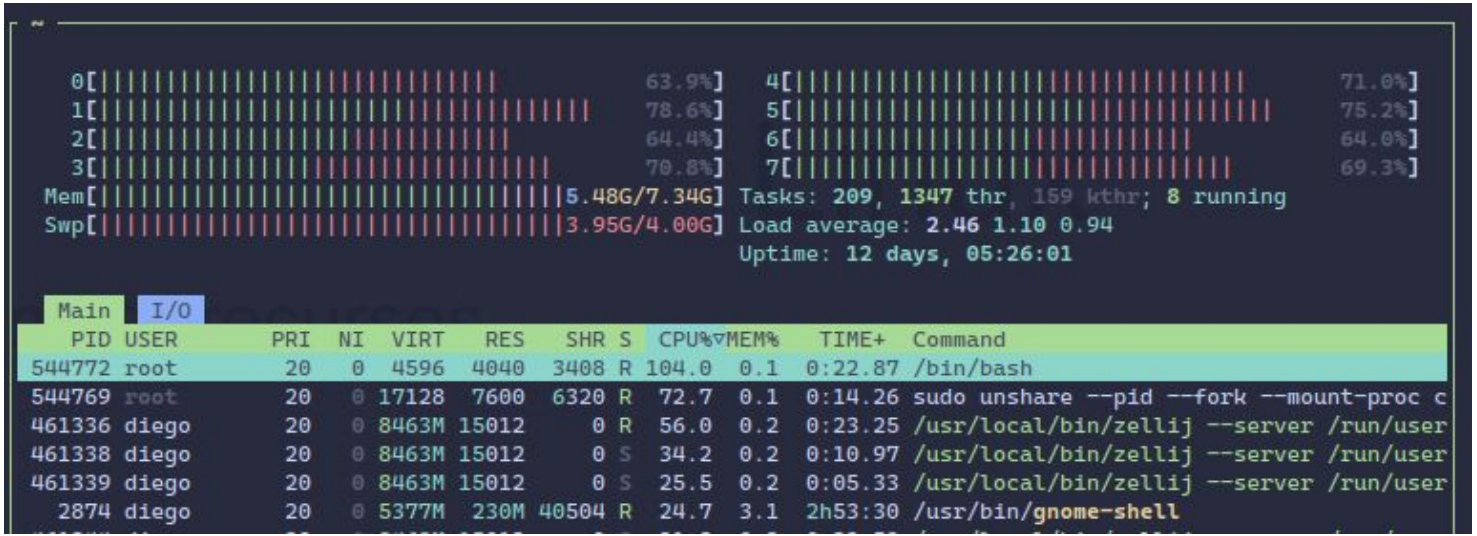
- Limitar recursos

```
~/container
ls /sys/fs/cgroup/
Permissions Size User Date Modified Name
drwxr-xr-x@    - root  4 mar 15:08  dev-hugepages
drwxr-xr-x@    - root  4 mar 15:08  dev-mqueue.mo
drwxr-xr-x     - root  4 mar 15:08  init.scope
drwxr-xr-x@    - root  4 mar 15:08  proc-sys-fs-b
drwxr-xr-x@    - root  4 mar 15:08  sys-fs-fuse-c
drwxr-xr-x@    - root  4 mar 15:08  sys-kernel-co
drwxr-xr-x@    - root  4 mar 15:08  sys-kernel-de
drwxr-xr-x@    - root  4 mar 15:08  sys-kernel-tr
drwxr-xr-x     - root  8 mar 14:40  system.slice
drwxr-xr-x@    - root  4 mar 15:08  user.slice
.r--r--r--     0 root  7 mar 19:06  cgroup.contro
.rw-r--r--     0 root  8 mar 14:44  cgroup.max.de
.rw-r--r--     0 root  8 mar 14:44  cgroup.max.de
.rw-r--r--     0 root  8 mar 14:44  cgroup.pressu
.rw-r--r--     0 root  8 mar 14:44  cgroup.procs
.r--r--r--     0 root  8 mar 14:44  cgroup.stat
.rw-r--r--     0 root  8 mar 07:06  cgroup.subtre
.rw-r--r--     0 root  8 mar 14:44  cgroup.thread
```

```
root@diego-XPS-13-9300: /sys/fs/cgroup/auladocker
root@diego-XPS-13-9300:.# echo "+cpu" >> /sys/fs/cgroup/cgroup.subtree_control
root@diego-XPS-13-9300:.# echo "+cpuset" >> /sys/fs/cgroup/cgroup.subtree_control
root@diego-XPS-13-9300:.# cd /sys/fs/cgroup/
chdir: erro ao obter o diretório atual: getcwd: impossível acessar os diretórios pa
 ou diretório inexistente
root@diego-XPS-13-9300:/sys/fs/cgroup# ls
cgroup.controllers      cpuset.cpus.isolated   io.cost.qos             misc.current
cgroup.max.depth        cpuset.mems.effective  io.pressure             misc.peak
cgroup.max.descendants  cpu.stat               io.prio.class           proc-sys-fs-
cgroup.pressure         cpu.stat.local         io.stat                 sys-fs-fuse-
cgroup.procs            dev-hugepages.mount    memory.numa_stat        sys-kernel-c
cgroup.stat             dev-mqueue.mount       memory.pressure         sys-kernel-c
cgroup.subtree_control  dmem.capacity          memory.reclaim          sys-kernel-t
cgroup.threads          dmem.current           memory.stat             system.slice
cpu.pressure            init.scope             memory.zswap.writeback  user.slice
cpuset.cpus.effective   io.cost.model          misc.capacity
root@diego-XPS-13-9300:/sys/fs/cgroup# mkdir auladocker/
root@diego-XPS-13-9300:/sys/fs/cgroup# cd auladocker/
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# ls
cgroup.controllers              dmem.current              memory.min
cgroup.events                   dmem.low                  memory.numa_stat
cgroup.freeze                   dmem.max                  memory.oom.group
cgroup.kill                     dmem.min                  memory.peak
cgroup.max.depth                hugetlb.1GB.current       memory.pressure
cgroup.max.descendants          hugetlb.1GB.events        memory.reclaim
cgroup.pressure                 hugetlb.1GB.events.local  memory.stat
cgroup.procs                    hugetlb.1GB.max           memory.swap.current
cgroup.stat                     hugetlb.1GB.numa_stat     memory.swap.events
```

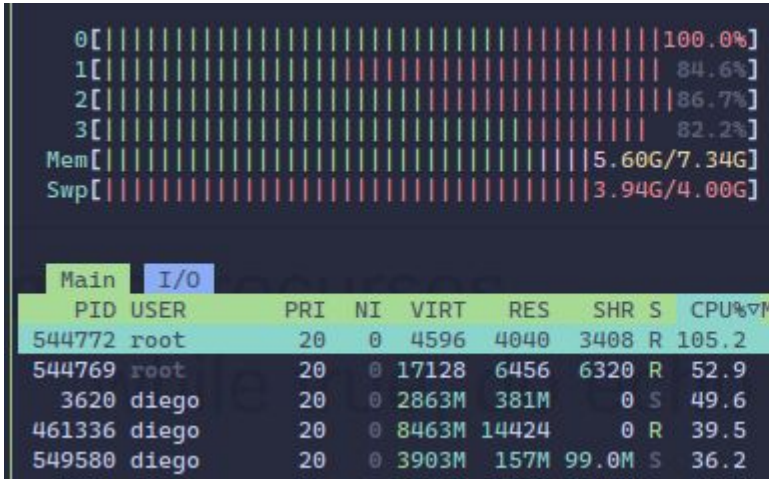
Linux

- Limitar recursos
  - while true; do echo aula; done



Linux

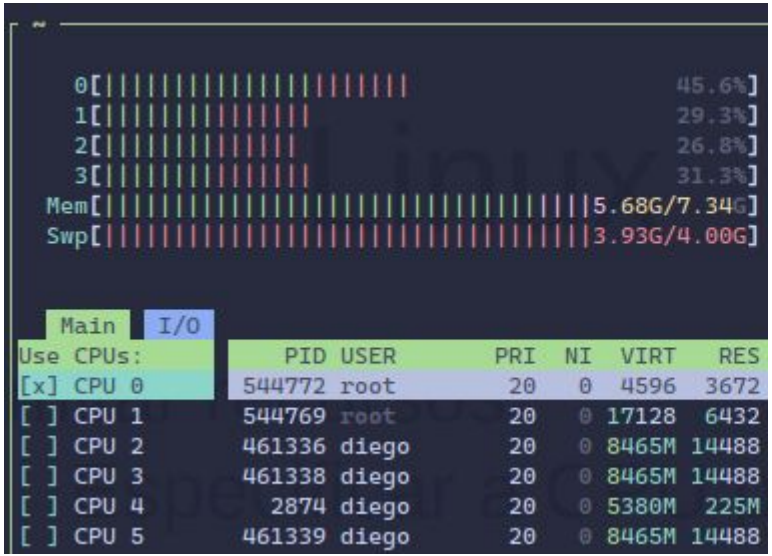
- Limitar recursos
  - Especificar a CPU para utilização




```
root@diego-XPS-13-9300: /sys/fs/cgroup/auladocker
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "+cpuset" >> /sys/fs/cgroup/auladocker/cgroup.subtre
e_control
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "+cpu" >> /sys/fs/cgroup/auladocker/cgroup.subtree_c
ontrol
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "544772" >> /sys/fs/cgroup/auladocker/cgroup.procs
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "o" >> /sys/fs/cgroup/auladocker/cpuset.cpus
bash: echo: erro de escrita: Argumento inválido
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "0" >> /sys/fs/cgroup/auladocker/cpuset.cpus
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker#
```

Linux

- Limitar recursos
  - Especificar a CPU para utilização
  - Limitar o consumo




```
root@diego-XPS-13-9300: /sys/fs/cgroup/auladocker
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "+cpuset" >> /sys/fs/cgroup/auladocker/cgroup.subtre
e_control
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "+cpu" >> /sys/fs/cgroup/auladocker/cgroup.subtree_c
ontrol
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "544772" >> /sys/fs/cgroup/auladocker/cgroup.procs
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "o" >> /sys/fs/cgroup/auladocker/cpuset.cpus
bash: echo: erro de escrita: Argumento inválido
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "0" >> /sys/fs/cgroup/auladocker/cpuset.cpus
root@diego-XPS-13-9300:/sys/fs/cgroup/auladocker# echo "500000 1000000" >> /sys/fs/cgroup/auladocker/cpu.ma
x
```

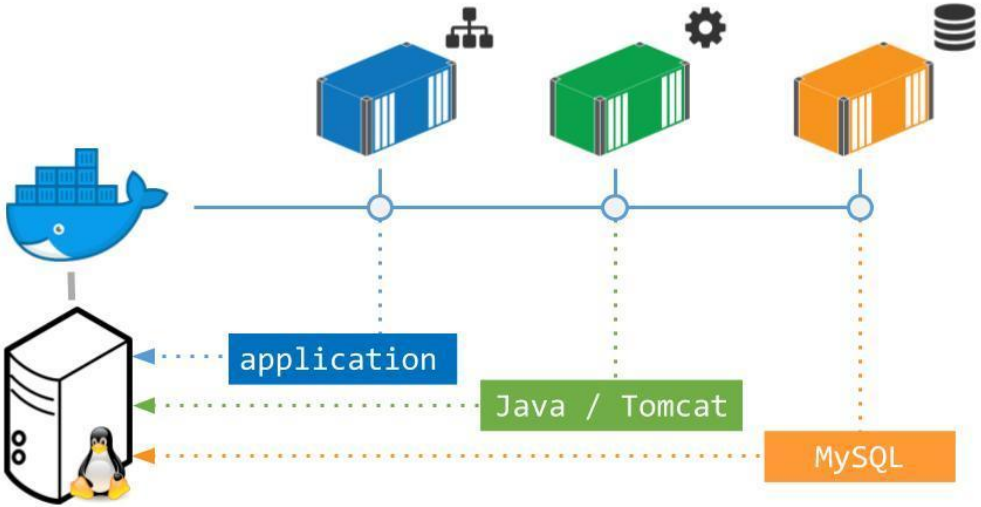
Linux

- Lxc
  - Criação de máquina virtuais dentro de containers
    - https://linuxcontainers.org
- Lxd
  - Ferramenta a partir do lxc
    - https://canonical.com/lxd

Docker

- Ambiente único
- Diversos ambiente
  - Idempotência
- Isolamento
- Gerenciamento de conflitos
- Segurança
  - Isolamento

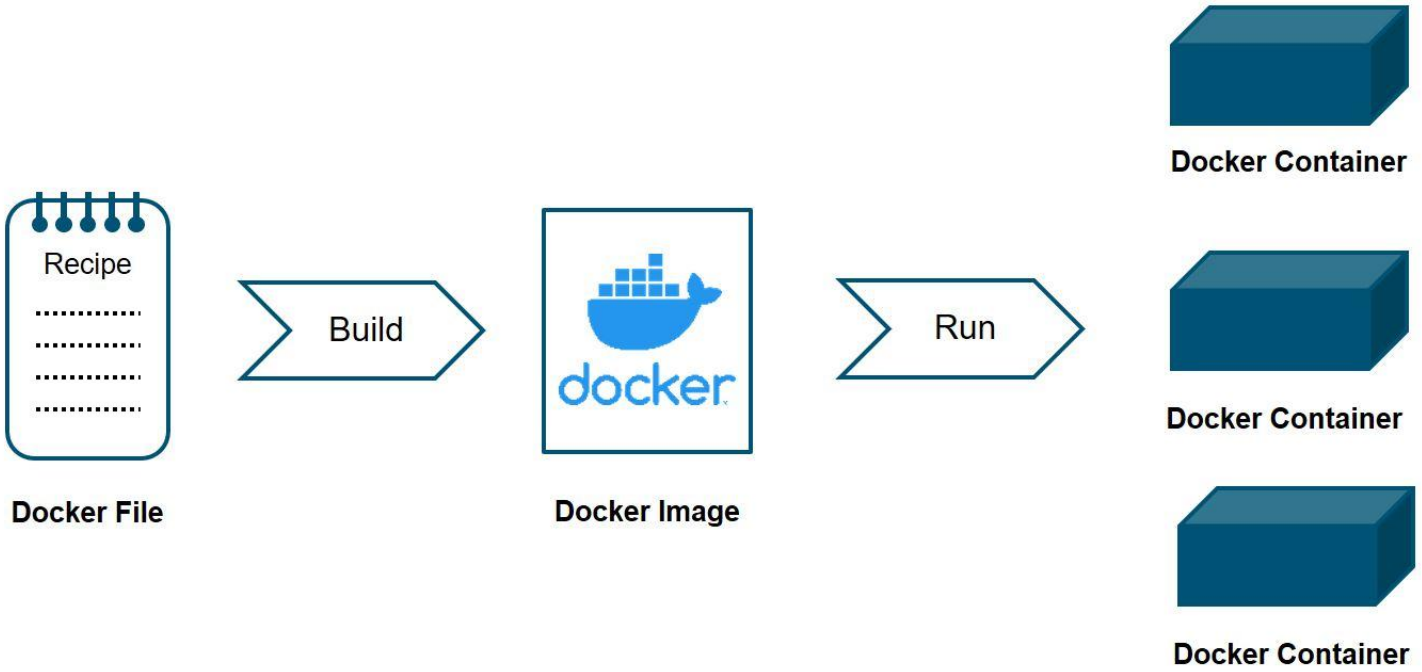
Agora irá rodar na máquina de todos!



Docker

- Imagem
  - Pacote com as dependências para criar um contêiner
  - Blueprint
- Dockerfile
  - Arquivo com as instruções sobre como criar uma imagem
- Build
  - Ação de criação de uma imagem

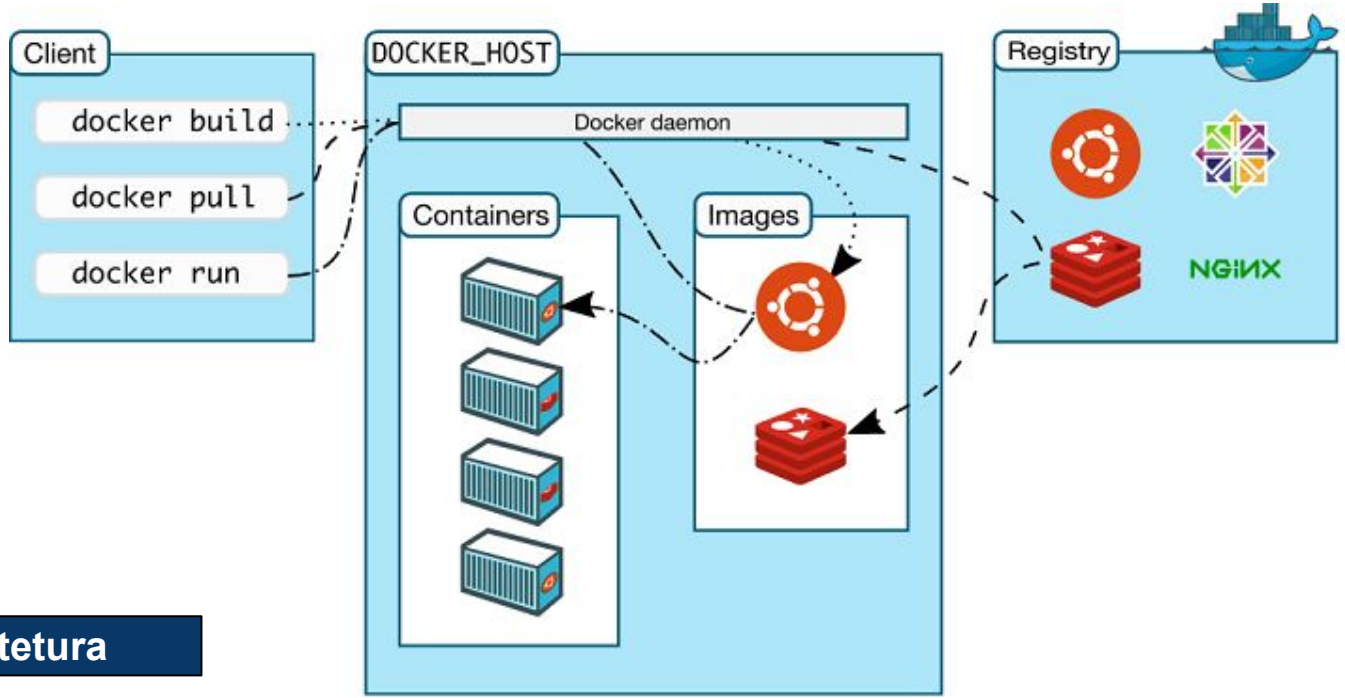
Docker


Recipe
Docker File
Build
docker
Docker Image
Run
Docker Container
Docker Container
Docker Container

Docker

- Contêiner
  - Instância de uma imagem
- Volume
  - Diretórios externos ao container
  - Persistir dados
- Tag
  - Nome que pode ser dado a uma imagem
  - Diferentes versões

Docker



Arquitetura

Container

- Antes de criar um projeto é necessário saber as necessidades do projeto
- Arquiteturas e linguagens diferentes possuem dependências e características diferentes
  - Composer
  - Npm
  - Maven
  - …

Container

- Instalar
  - https://docs.docker.com/engine/install
- Testando
  - docker container run hello-world
  - docker run hello-world
  - docker container run --name teste hello-world
- Listar container
  - docker container ls (Somente ativos)
  - docker container ls -a

```
root@diego:/home/diego# docker container ls -a
CONTAINER ID   IMAGE         COMMAND    CREATED         STATUS                     PORTS     NAMES
dcf9a79556d6   hello-world   "/hello"   2 minutes ago   Exited (0) 2 minutes ago             teste
```

Container

- Remover
  - docker container rm teste
  - docker container rm dcf9a79556d6
  - docker container rm -f dcf9a79556d6
  - docker container run --name teste2 --rm hello-world
- Container interativo tty
  - docker container run -it ubuntu /bin/bash

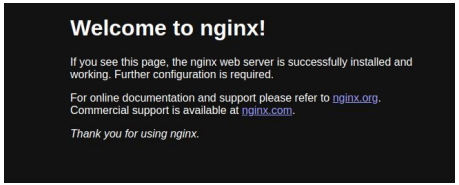
```
root@diego:/home/diego# docker container run -it ubuntu /bin/bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
01007420e9b0: Pull complete
Digest: sha256:f9d633ff6640178c2d0525017174a688e2c1aef28f0a0130b26bd5554491f0da
Status: Downloaded newer image for ubuntu:latest
root@78f7e9a85212:/#
```

Container

- Contínuo
  - Isolado
    - docker container run nginx
    - docker container run -d nginx
  - Bind
    - docker container run -d -p 8080:80 nginx

Local

container




```
root@diego:/home/diego# docker container run -d -p 8080:80 nginx
30efb3a713f17fd0455385c49565ce136fae0645c890c1c40e53041fd2a66233
root@diego:/home/diego# docker container ls -a
CONTAINER ID   IMAGE         COMMAND                  CREATED          STATUS                      PORTS                                   NAMES
30efb3a713f1   nginx         "/docker-entrypoint.…"   7 seconds ago    Up 6 seconds                0.0.0.0:8080->80/tcp, :::8080->80/tcp   musing_dijkstra
78f399c82381   nginx         "/docker-entrypoint.…"   5 minutes ago    Exited (0) 11 seconds ago                                           amazing_tharp
```

Container

- Variáveis de ambiente

```
docker run -d
-p 3306:3306
--name mariadb
-e MARIADB_USER=teste
-e MARIADB_ROOT_PASSWORD=teste
-e MARIADB_DATABASE=teste
mariadb:latest
```

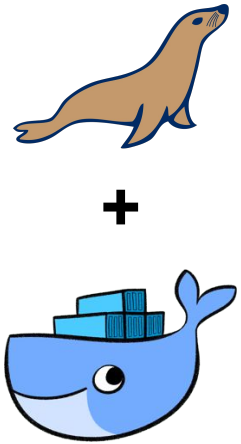
Isolado

Porta

Nome para referenciar

Parâmetros

Imagem



```
root@diego:/home/diego# docker container ls -a
CONTAINER ID   IMAGE           COMMAND                  CREATED         STATUS         PORTS                                       NAMES
6a3342f0d50d   mariadb:latest  "docker-entrypoint.s…"   9 seconds ago   Up 8 seconds   0.0.0.0:3306->3306/tcp, :::3306->3306/tcp   mariadb
```

Container

- Iniciar e parar
  - docker container start 6a3342f0d50d
  - docker container stop 6a3342f0d50d

- Inspecionar
  - docker container inspect 6a3342f0d50d
  - docker container exec -it 6a3342f0d50d /bin/bash
  - docker container logs 6a3342f0d50d -n -10
  - docker container logs 6a3342f0d50d -f -t

Imagem

- Blueprint do sistema a ser inicializado
- Possui todas as necessidades para um sistemas
- OverlayFs [Linux]
  - Cria um sistema de arquivo baseado em outro sistema de arquivo
  - Empilhamento de camadas
    - Sempre acumula, nunca escreve por cima

Imagem

```
sudo mkdir -p /overlay/primeira_camada
[sudo] senha para diego:
sudo mkdir -p /overlay/segunda_camada
sudo mkdir -p /overlay/work
sudo mkdir -p /overlay/merge
```

```
tree /overlay/
/overlay/
├── merge
├── primeira_camada
├── segunda_camada
└── work
```

```
sudo echo "arquivo 1" | sudo tee /overlay/primeira_camada/01_camada.txt
arquivo 1
sudo echo "arquivo 1" | sudo tee /overlay/segunda_camada/02_camada.txt
arquivo 1
tree /overlay/
/overlay/
├── merge
├── primeira_camada
│   └── 01_camada.txt
├── segunda_camada
│   └── 02_camada.txt
└── work

5 directories, 2 files
```

Imagem

```
sudo mount -t overlay overlay \
-o lowerdir=/overlay/primeira_camada,upperdir=/overlay/segunda_camada,workdir=/overlay/work \
/overlay/merge
tree /overlay/
/overlay/
├── merge
│   ├── 01_camada.txt
│   └── 02_camada.txt
├── primeira_camada
│   └── 01_camada.txt
├── segunda_camada
│   └── 02_camada.txt
└── work
    └── work  [error opening dir]

6 directories, 4 files
```

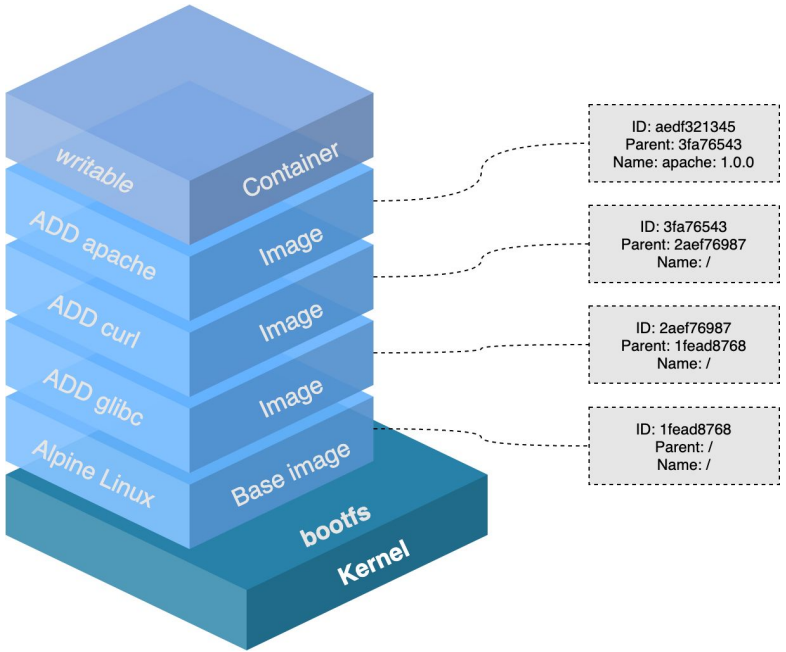


```
sudo echo "merge" | sudo tee /overlay/merge/merge.txt
merge
tree /overlay/
/overlay/
├── merge
│   ├── 01_camada.txt
│   ├── 02_camada.txt
│   └── merge.txt
├── primeira_camada
│   └── 01_camada.txt
├── segunda_camada
│   ├── 02_camada.txt
│   └── merge.txt
└── work
    └── work  [error opening dir]

6 directories, 6 files
```

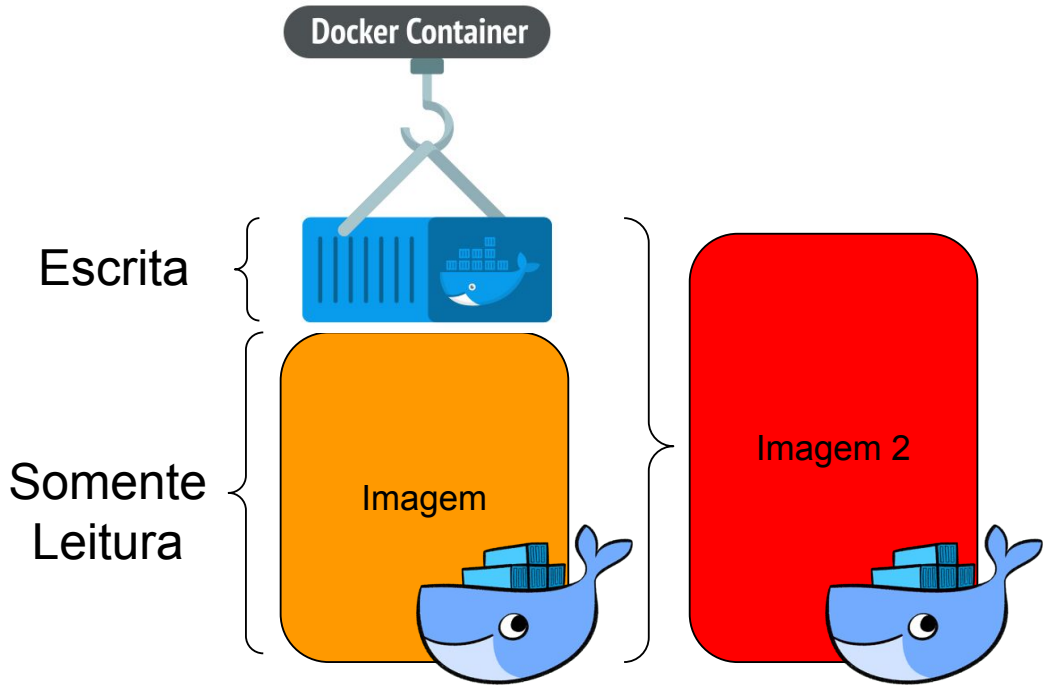
Imagem

- Construir
  - Docker commit
  - Dockerfile
    - Automatizar
    - Reproduzir
    - Versionar



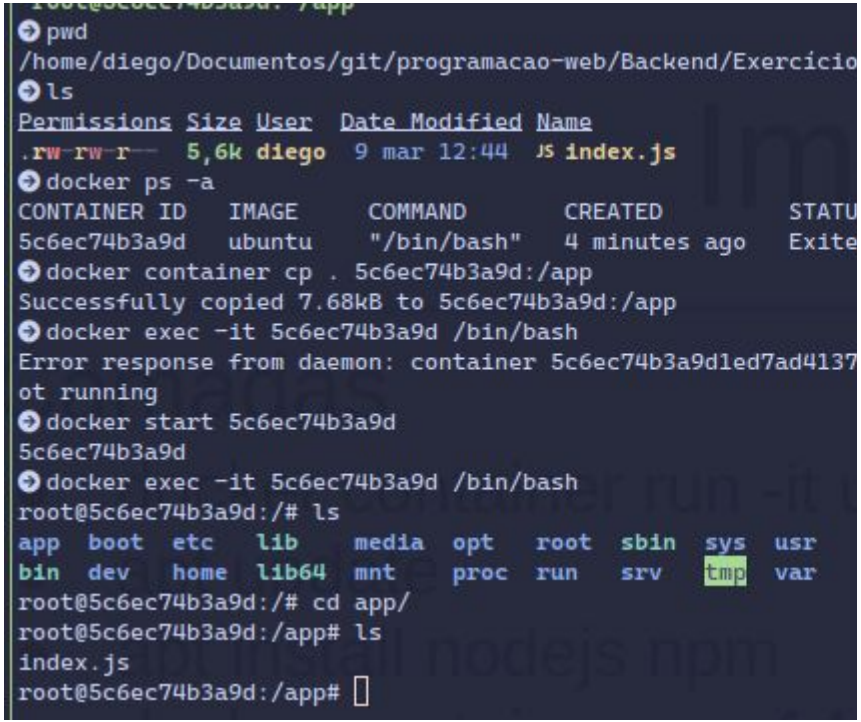
Imagem

- Docker commit
  - Junta o container atual com a imagem e gera uma nova imagem



Imagem

- Camadas
  - docker container run -it ubuntu /bin/bash
  - apt update
  - apt install nodejs npm
  - docker container cp . 14d14531b90f



Imagem

- Camadas
  - docker container run -it ubuntu /bin/bash
  - apt update
  - apt install nodejs npm
  - docker container cp . 14d14531b90f

```
docker commit 5c6ec74b3a9d primeira_imagem
sha256:9d0b90626424b2395b94771b984730212a0092d641e65c6b0ded31623fa65adb
docker image ls
REPOSITORY        TAG       IMAGE ID       CREATED          SIZE
primeira_imagem   latest    9d0b90626424   13 seconds ago   233MB
ubuntu            latest    bbdabce66f1b   3 weeks ago      78.1MB
```

```
docker history primeira_imagem
IMAGE          CREATED         CREATED BY                                      SIZE
9d0b90626424   3 minutes ago   /bin/bash                                       155MB
bbdabce66f1b   3 weeks ago     /bin/sh -c #(nop)  CMD ["/bin/bash"]            0B
<missing>      3 weeks ago     /bin/sh -c #(nop) ADD file:1ae27d2ef43693611…   78.1MB
<missing>      3 weeks ago     /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      3 weeks ago     /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      3 weeks ago     /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH     0B
<missing>      3 weeks ago     /bin/sh -c #(nop)  ARG RELEASE                  0B
```

Imagem

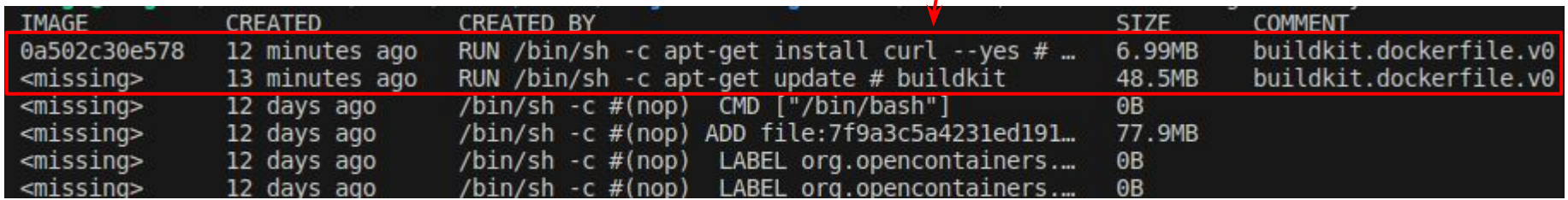

```
        "Id": "sha256:9d0b90626424b2395b94771b984730212a0092d641e65c6b0ded31623fa65adb",
        "RepoTags": [
            "primeira_imagem:latest"
        ],
        "RepoDigests": [],
        "Parent": "sha256:bbdabce66f1b7dde0c081a6b4536d837cd81dd322dd8c99edd68860baf3b2db3",
        "Comment": "",
        "Created": "2026-03-09T17:16:54.362077264Z",
        "DockerVersion": "28.4.0",
        "Author": "",
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 233139315,
        "GraphDriver": {
            "Data": {
                "LowerDir": "/var/lib/docker/overlay2/50c081bdb48f9eaa4546f9ff3e1472fd057e2de20001ef8c194e4
154f43ad8ef/diff",
                "MergedDir": "/var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f
0356fc2f7511/merged",
                "UpperDir": "/var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f0
356fc2f7511/diff",
                "WorkDir": "/var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f03
56fc2f7511/work"
```

Imagem

```
root@diego-XPS-13-9300: /var/lib/docker/overlay2/b[...]df91f2a5dc414537292130f66c3659c9f0356fc2f7511/diff
root@diego-XPS-13-9300:/# cd /var/lib/docker/overlay2/50c081bdb48f9eaa4546f9ff3e1472fd057e2de20001ef8c194e4
154f43ad8ef/diff
root@diego-XPS-13-9300:/var/lib/docker/overlay2/50c081bdb48f9eaa4546f9ff3e1472fd057e2de20001ef8c194e4154f43
ad8ef/diff# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@diego-XPS-13-9300:/var/lib/docker/overlay2/50c081bdb48f9eaa4546f9ff3e1472fd057e2de20001ef8c194e4154f43
ad8ef/diff# cd /var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f0356fc2f7511/di
ff
root@diego-XPS-13-9300:/var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f0356fc2
f7511/diff# ls
app  etc  root  tmp  usr  var
root@diego-XPS-13-9300:/var/lib/docker/overlay2/b4720db6d19a462e9fcdf91f2a5dc414537292130f66c3659c9f0356fc2
f7511/diff#
```

Imagem

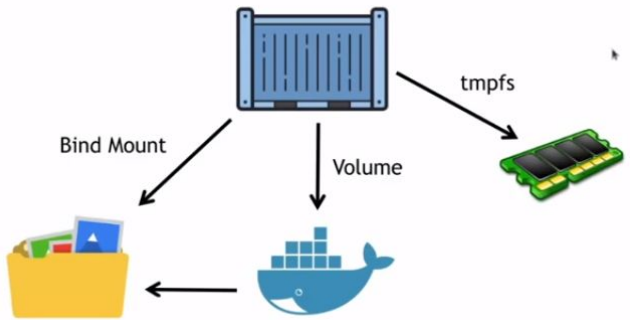
- Outros comandos
  - docker image prune
    - docker container rm -f ${docker container ls -qa}
  - docker image rm nginx
  - docker image inspect nginx
  - docker image history nginx



Volume

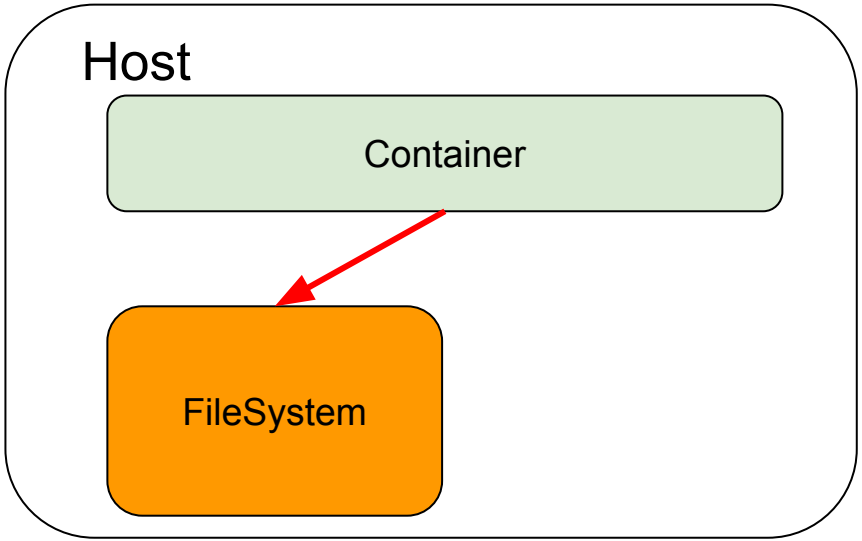
- Stateless X Statefull
- Container ephemeral
  - Perda de dados
  - Banco de dados
- Mapeamento de um diretório na sua máquina para armazenar as informações que estavam dentro do container

- Tipos
  - Bind mount
  - Volume
  - Tmpfs



Volume

- Bind mount
  - Vincula uma pasta do container direto com uma pasta do seu SO
  - Cuidado com os usuários da máquina real

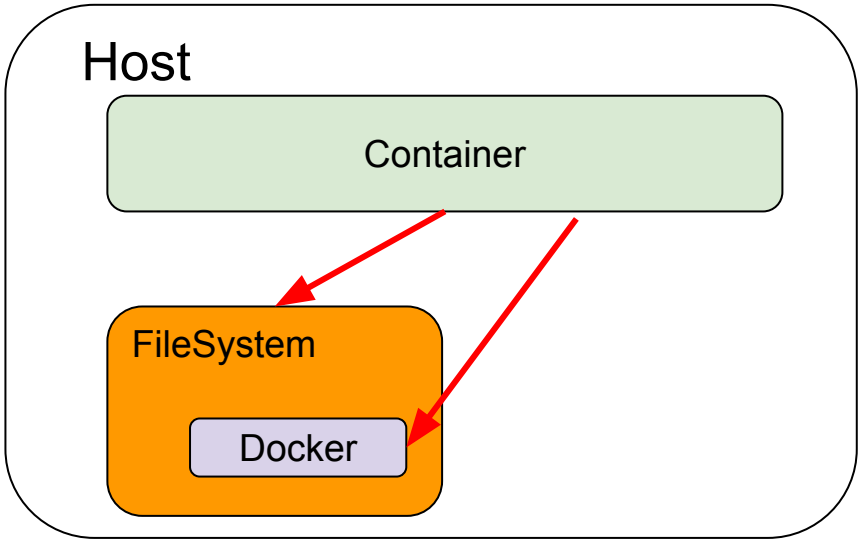



```
docker container run -it --mount
type=bind, src="$(pwd)", target=/app
ubuntu /bin/bash
```

```
docker container run -it -v "$(pwd)":/app
ubuntu /bin/bash
```

Volume

- Gerenciado pelo docker
  - Named / docker volume
  - docker volume create volume_cefet
  - docker volume ls
  - docker volume rm volume_cefet



docker container run -it --mount type=volume, src=volume_cefet, target=/app ubuntu /bin/bash

docker container run -it -v volume_cefet ubuntu /bin/bash

Volume

- Gerenciado pelo docker
  - docker volume inspect volume_cefet
    - Não é para acessar! Para não ter o problemas do bind

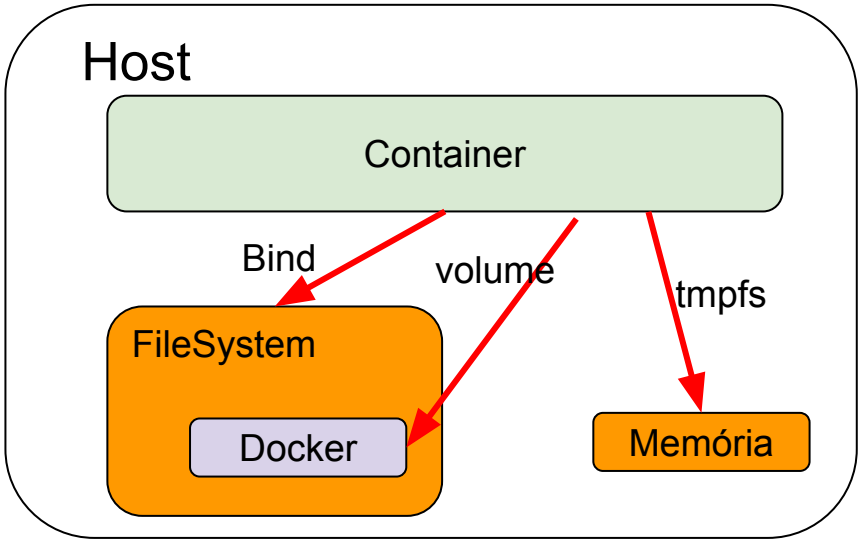

```
[
    {
        "CreatedAt": "2023-11-24T18:52:08-03:00",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/aula_vol/_data",
        "Name": "aula_vol",
        "Options": null,
        "Scope": "local"
    }
]
```

Volume

- Backup
  - docker container ls -a
  - docker container run --volumes-from id --rm -v ${pwd}:/backup ubuntu:22.04 tar cvf /backup/backup_vol.tar /app
- Restore
  - docker container run -v $(pwd):/backup -v novo_volume:/app ubuntu:22.04 tar xvf /backup/backup_vol.tar

Volume

- tmpfs
  - Fica na memória
  - Os dados serão perdidos
  - Será mais rápido




```
docker container run -it --mount
type=tmpfs, target=/app ubuntu /bin/bash
```

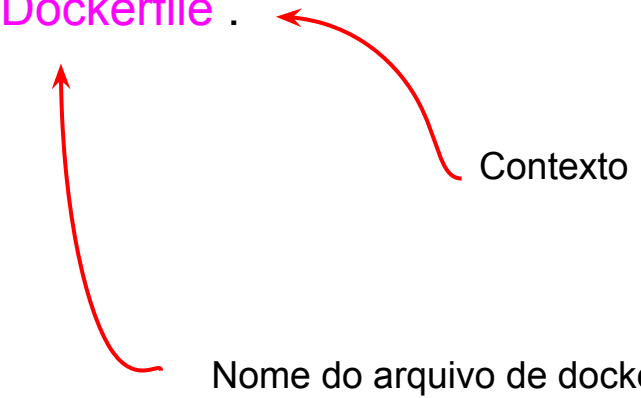
```
docker container run -it --tmpfs /app
ubuntu /bin/bash
```

Dockerfile

- Conjunto de passos necessário para montar uma imagem
- Todo container tem como base uma imagem
  - Toda imagem tem como base uma imagem
  - Para isso usamos o FROM
- RUN vai rodar o shell

Dockerfile

- Construindo
  - docker image build -t nome_da_imagem -f Dockerfile .

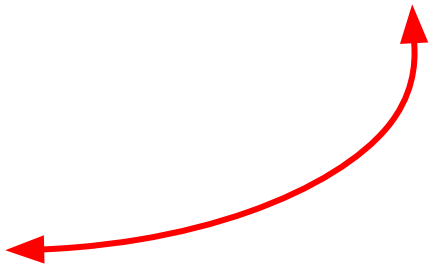



```
1  FROM ubuntu
2  RUN apt-get update
3  RUN apt-get install curl --yes
4
```

Dockerfile

- Cache
  - docker image build -t nome_da_imagem .
  - Ordem compromete o cache
  - Sem cache de camada
    - -- no-cache

```
1  FROM ubuntu
2  RUN apt-get update
3  RUN apt-get install curl --yes
4
```



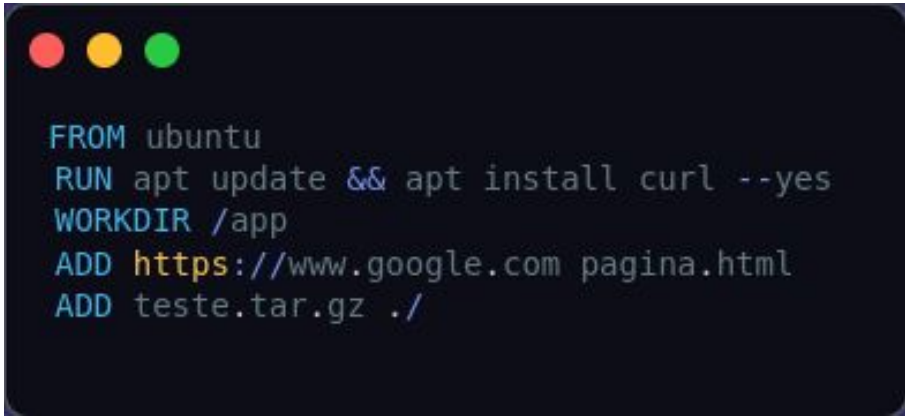
```
1  FROM ubuntu
2  RUN apt-get update && apt-get install curl --yes
3
```

Dockerfile

- FROM => Inicializa o build de uma imagem a partir de uma imagem base
- RUN => Executa um comando
- WORKDIR => Define o seu diretório corrente
  - Basicamente faz um mkdir e um cd
- COPY => Copia arquivos ou diretórios e adiciona ao sistema de arquivos da imagem

Dockerfile

- ADD => Copia arquivos ou diretórios ou arquivos remotos e adiciona ao sistema de arquivos da imagem
  - Não usa cache



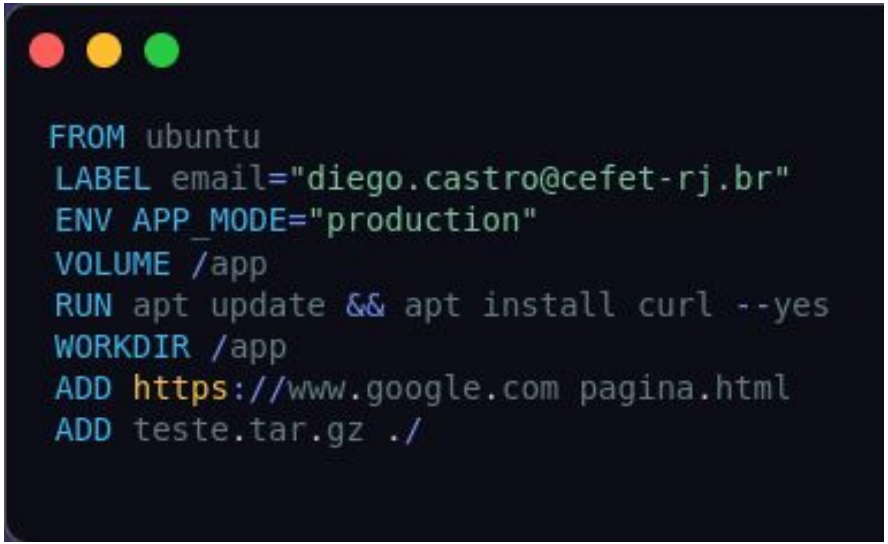
Dockerfile

- LABEL => Adiciona metadados a imagem
  - Não cria nova camada
- ENV => Define variáveis de ambiente
  - --env APP_MODE = “Dev”



Dockerfile

- VOLUME => Define volumes que devem ser definidos
  - Deixa de ser efêmero



```
FROM ubuntu
LABEL email="diego.castro@cefet-rj.br"
ENV APP_MODE="production"
VOLUME /app
RUN apt update && apt install curl --yes
WORKDIR /app
ADD https://www.google.com pagina.html
ADD teste.tar.gz ./
```

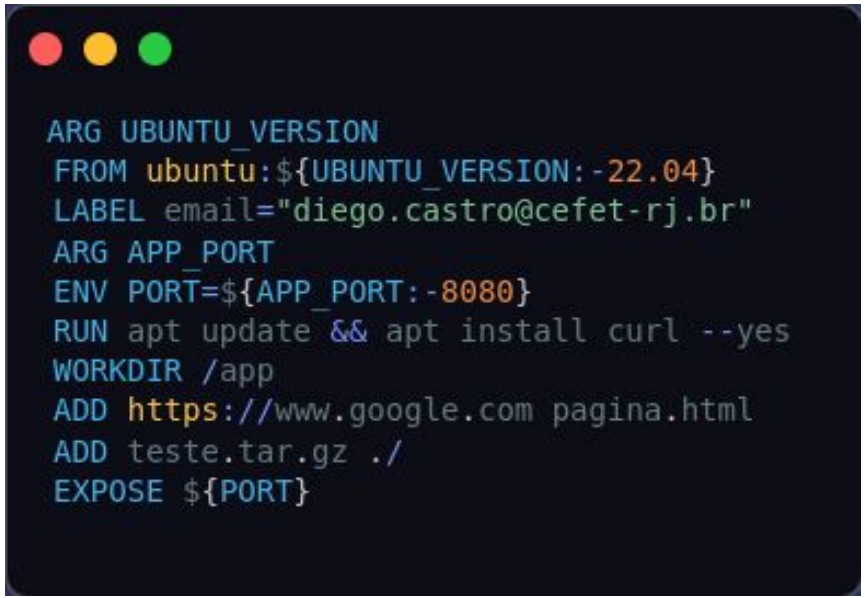
Dockerfile

- ARG => Define um argumento pra ser usado no processo de construção
  - docker build -t ubuntu --build -arg Version=22.04 .

```
ARG UBUNTU_VERSION=22.04
FROM ubuntu:${UBUNTU_VERSION}
LABEL email="diego.castro@cefet-rj.br"
ARG CURL_VERSION=7.81.0
ENV APP_MODE="production"
RUN apt update && apt install curl=${CURL_VERSION}* --yes
WORKDIR /app
ADD https://www.google.com pagina.html
ADD teste.tar.gz ./
```

Dockerfile

- EXPOSE => Define que o container precisa expor a porta em questão
  - -p 8080:80
  - -P



```
ARG UBUNTU_VERSION
FROM ubuntu:${UBUNTU_VERSION:-22.04}
LABEL email="diego.castro@cefet-rj.br"
ARG APP_PORT
ENV PORT=${APP_PORT:-8080}
RUN apt update && apt install curl --yes
WORKDIR /app
ADD https://www.google.com pagina.html
ADD teste.tar.gz ./
EXPOSE ${PORT}
```

Dockerfile

- USER => Define o Usuário que vai ser usado
  - Não usar o root

```
ARG UBUNTU_VERSION
FROM ubuntu:${UBUNTU_VERSION:-22.04}
LABEL email="diego.castro@cefet-rj.br"
ARG APP_PORT
ENV PORT=${APP_PORT:-8080}
RUN apt update && apt install curl --yes
WORKDIR /app

RUN useradd diego_user
ADD --chown=diego_user:diego_user https://www.google.com pagina.html
ADD --chown=diego_user:diego_user teste.tar.gz ./

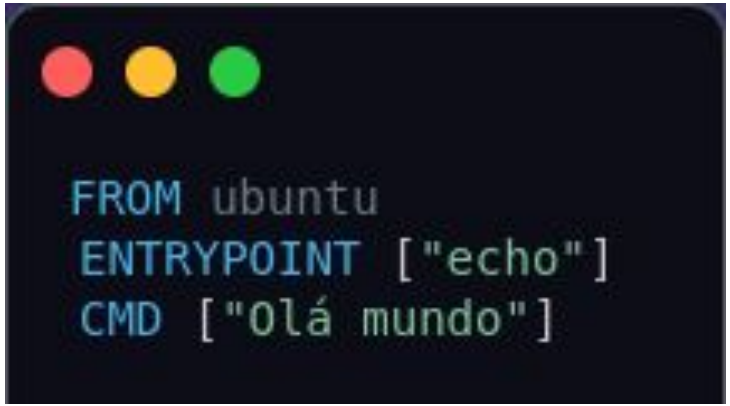
USER diego_user

EXPOSE ${PORT}
```

Dockerfile

- CMD => Define o comando e/ou os parâmetros padrão
  - Comando padrão que será executado quando o container iniciar
  - Usado para substituição
- ENTRYPOINT => Ajuda você a configurar um contêiner que pode ser executado como um executável
  - Comando fixo que sempre será executado
  - --entrypoint

Dockerfile



- docker run minha-imagem
  - Olá mundo
- docker run minha-imagem Diego
  - Diego

Dockerfile

- entrypoint.sh
  - Muito utilizado para configurar ambientes mais complexos

```
#!/bin/bash
if [ -z $1 ]
then
    echo "Iniciando sem parametro"
else
    echo "Iniciando com o parametro $1"
fi
```

```
FROM ubuntu
WORKDIR /app
COPY --chown=root:root --chmod=100 ./entrypoint.sh .
ENTRYPOINT ["./entrypoint.sh"]
```

Dockerfile

```
const express = require("express");
// import express from "express";

var xmlparser = require('express-xml-bodyparser');

const app = express();
//app.use(express.json());
app.use(xmlparser())

app.post("/aluno", (request, response) => {

    console.log(request.body);

    return response.json(
        { "nome": "aluno 4" }
    )
})

app.get("/alunos", (request, response) => {
    const query = request.query;
    const { pagina, count } = request.query;

    console.log("request.query.count: " + request.query.count);
    console.log("query.pagina: " + query.pagina);
    console.log("pagina: " + pagina);
    console.log("count: " + count);

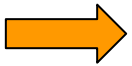

    return response.json([
        { "nome": "aluno 1" },
        { "nome": "aluno 2" },
        { "nome": "aluno 3" },
    ])
})
```



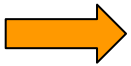
```
FROM node
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
EXPOSE 3000
CMD ["node", "src/index"]
```

Boas práticas

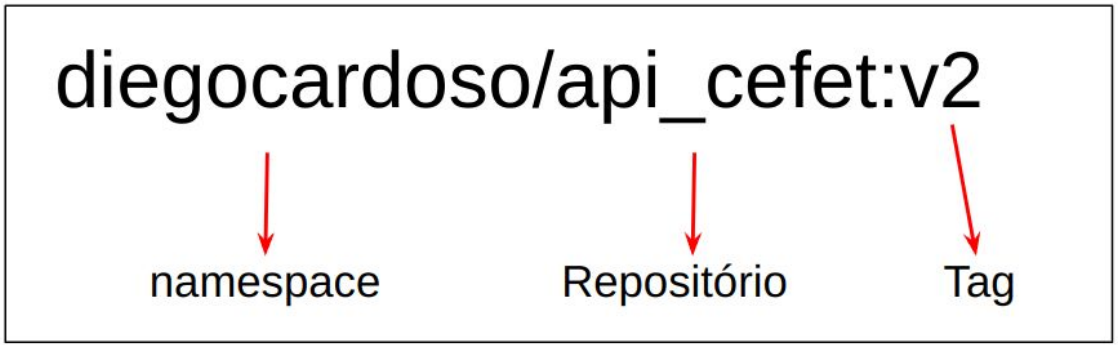
```dockerfile
FROM joaozinho/node
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
EXPOSE 3000
CMD ["node", "src/index"]
```



```dockerfile
FROM node
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
EXPOSE 3000
CMD ["node", "src/index"]
```



```dockerfile
FROM node:21
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
EXPOSE 3000
CMD ["node", "src/index"]
```



Boas práticas

- Docker ignore
- FROM node:21 X FROM node:21-alpine
- CMD X ENTRYPOINT
  - CMD ["node", "src/index"]
  - ENTRYPOINT -> arquivo .sh

.dockerignore

```
1  node_modules
```

```
1  FROM ubuntu
2  WORKDIR /app
3  COPY ./entrypoint.sh ./entrypoint.sh
4  RUN chmod +x ./entrypoint.sh
5  CMD ["./entrypoint.sh"]
```

```
1  #!/bin/bash
2  echo "Inicinado o container"
```

```
root@diego:/home/diego/Documentos/express-aula1# docker container run --name ubuntu_echo ubuntu
Container subindo!
root@diego:/home/diego/Documentos/express-aula1# docker container run --name ubuntu_echo2 ubuntu echo "Subindo container"
Subindo container
root@diego:/home/diego/Documentos/express-aula1#
```

Boas práticas

- Argumentos
  - docker image build
    -t ubuntu-arg:V1
    --build-arg TAG="21" .

```
1  ARG TAG=latest
2  FROM node:$TAG
3  WORKDIR /app
4  COPY package*.json ./
5  RUN npm i
6  COPY . .
7  EXPOSE 3000
8  CMD ["node", "src/index"]
```

- Multistage Build
  - Interpretada
    - PHP, Javascript
  - Compiladas
    - Go, C, C++
  - JIT(Just in time)
    - Java, PHP > 8

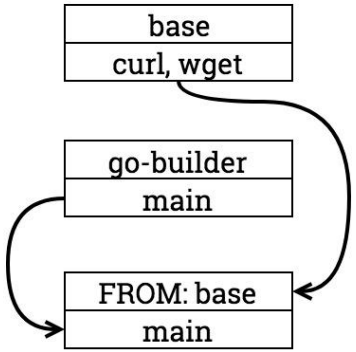
Boas práticas

- Multistage Build
  - Imagens intermediárias

Compila

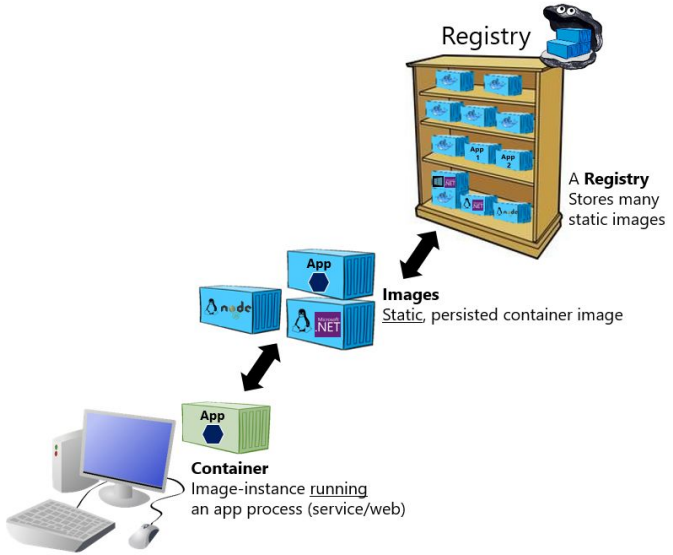
Executa Final

```
1   FROM alpine AS base
2   RUN apk add --no-cache curl wget
3
4   FROM golang:1.9.2 AS go-builder
5   WORKDIR /go
6   COPY *.go /go/
7   RUN CGO_ENABLED=0 GOOS=linux GOARCH=amd64 go build -o main .
8
9   FROM base
10  COPY --from=go-builder /go/main /main
11  CMD ["/main"]
```



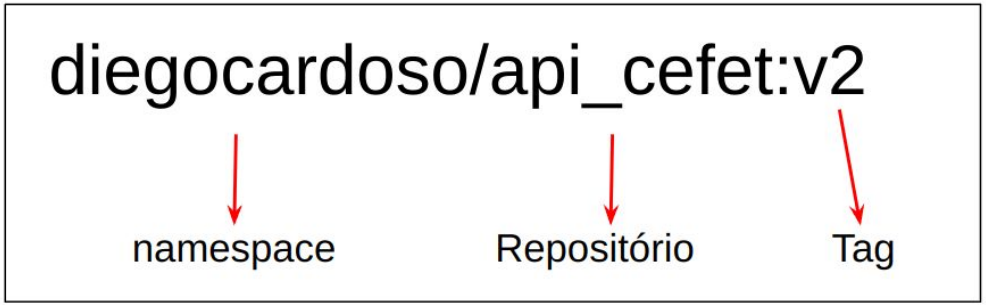
Docker Registry

- Repositório de imagens
  - Públicas / privadas
- Exemplos
  - Docker hub
  - Elastic Container Registry
  - Azure Container Registry
  - Google Container Registry
  - Harbor



Docker Registry

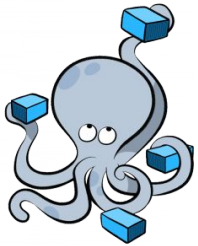
- Namespace
- Docker hub
  - docker login
    - usuário
    - senha
  - docker commit namespace/container:tag
  - docker push namespace/container:tag
  - docker tag nome/tag_antiga nome/tag_nova



Network

- Vários processos no mesmo container 
- Redes
  - Bridge
    - Comunicação entre containers
    - Port Bind
  - Host
    - Remove isolamento entre o container e sua máquina
  - Overlay
    - Comunica vários docker hosts
  - Macvlan
    - Simula um endereço mac

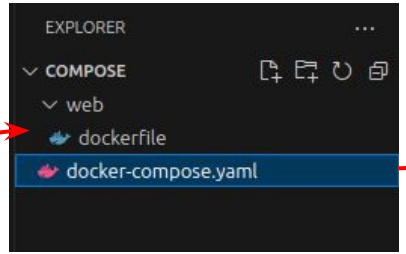
Docker Compose



- Receita de criação de containers
- Dockerfile -> imagens docker
- Docker Compose -> criação de containe
- docker-compose.yaml
- Duas formas de fazer

Dockerfile

```
ARG TAG=latest
FROM node:$TAG
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
EXPOSE 3000
CMD ["node", "src/index"]
```




```
version: "3.8"

networks:
  subnet:
    driver: bridge

services:
  db:
    image: postgres:15-alpine
    restart: always
    environment:
      POSTGRES_PASSWORD: postgres
      POSTGRES_DB: postgres
      POSTGRES_USER: postgres
    ports:
      - "15432:5432"
    networks:
      - subnet

  web:
    build: ./web
    environment:
      MYSQL_DATABASE: test
      MYSQL_USER: root
      MYSQL_PASSWORD: password
      MYSQL_HOST: db
    ports:
      - "5000:5000"
    networks:
      - subnet
    depends_on:
      - db
    restart: on-failure
```